

Iris dataset

Classify iris plants into three species in this classic dataset

sepal_length	sepal_width	petal_length	petal_width	species
5.1	3.5	1.4	0.2	setosa
4.9	3	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa
4.6	3.4	1.4	0.3	setosa
5	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa
5.4	3.7	1.5	0.2	setosa
4.8	3.4	1.6	0.2	setosa
4.8	3	1.4	0.1	setosa
4.3	3	1.1	0.1	setosa
5.8	4	1.2	0.2	setosa
5.7	4.4	1.5	0.4	setosa
5.4	3.9	1.3	0.4	setosa
5.1	3.5	1.4	0.3	setosa
5.7	3.8	1.7	0.3	setosa
5.1	3.8	1.5	0.3	setosa
5.4	3.4	1.7	0.2	setosa
5.1	3.7	1.5	0.4	setosa
4.6	3.6	1	0.2	setosa
5.1	3.3	1.7	0.5	setosa
4.8	3.4	1.9	0.2	setosa
5	3	1.6	0.2	setosa
5	3.4	1.6	0.4	setosa
5.2	3.5	1.5	0.2	setosa
5.2	3.4	1.4	0.2	setosa
4.7	3.2	1.6	0.2	setosa
4.8	3.1	1.6	0.2	setosa
5.4	3.4	1.5	0.4	setosa
5.2	4.1	1.5	0.1	setosa
5.5	4.2	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa
5	3.2	1.2	0.2	setosa
5.5	3.5	1.3	0.2	setosa
4.9	3.1	1.5	0.1	setosa

4.4	3	1.3	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5	3.5	1.3	0.3	setosa
4.5	2.3	1.3	0.3	setosa
4.4	3.2	1.3	0.2	setosa
5	3.5	1.6	0.6	setosa
5.1	3.8	1.9	0.4	setosa
4.8	3	1.4	0.3	setosa
5.1	3.8	1.6	0.2	setosa
4.6	3.2	1.4	0.2	setosa
5.3	3.7	1.5	0.2	setosa
5	3.3	1.4	0.2	setosa
7	3.2	4.7	1.4	versicolor
6.4	3.2	4.5	1.5	versicolor
6.9	3.1	4.9	1.5	versicolor
5.5	2.3	4	1.3	versicolor
6.5	2.8	4.6	1.5	versicolor
5.7	2.8	4.5	1.3	versicolor
6.3	3.3	4.7	1.6	versicolor
4.9	2.4	3.3	1	versicolor
6.6	2.9	4.6	1.3	versicolor
5.2	2.7	3.9	1.4	versicolor
5	2	3.5	1	versicolor
5.9	3	4.2	1.5	versicolor
6	2.2	4	1	versicolor
6.1	2.9	4.7	1.4	versicolor
5.6	2.9	3.6	1.3	versicolor
6.7	3.1	4.4	1.4	versicolor
5.6	3	4.5	1.5	versicolor
5.8	2.7	4.1	1	versicolor
6.2	2.2	4.5	1.5	versicolor
5.6	2.5	3.9	1.1	versicolor
5.9	3.2	4.8	1.8	versicolor
6.1	2.8	4	1.3	versicolor
6.3	2.5	4.9	1.5	versicolor
6.1	2.8	4.7	1.2	versicolor
6.4	2.9	4.3	1.3	versicolor
6.6	3	4.4	1.4	versicolor
6.8	2.8	4.8	1.4	versicolor
6.7	3	5	1.7	versicolor
6	2.9	4.5	1.5	versicolor
5.7	2.6	3.5	1	versicolor
5.5	2.4	3.8	1.1	versicolor
5.5	2.4	3.7	1	versicolor
5.8	2.7	3.9	1.2	versicolor

6	2.7	5.1	1.6	versicolor
5.4	3	4.5	1.5	versicolor
6	3.4	4.5	1.6	versicolor
6.7	3.1	4.7	1.5	versicolor
6.3	2.3	4.4	1.3	versicolor
5.6	3	4.1	1.3	versicolor
5.5	2.5	4	1.3	versicolor
5.5	2.6	4.4	1.2	versicolor
6.1	3	4.6	1.4	versicolor
5.8	2.6	4	1.2	versicolor
5	2.3	3.3	1	versicolor
5.6	2.7	4.2	1.3	versicolor
5.7	3	4.2	1.2	versicolor
5.7	2.9	4.2	1.3	versicolor
6.2	2.9	4.3	1.3	versicolor
5.1	2.5	3	1.1	versicolor
5.7	2.8	4.1	1.3	versicolor
6.3	3.3	6	2.5	virginica
5.8	2.7	5.1	1.9	virginica
7.1	3	5.9	2.1	virginica
6.3	2.9	5.6	1.8	virginica
6.5	3	5.8	2.2	virginica
7.6	3	6.6	2.1	virginica
4.9	2.5	4.5	1.7	virginica
7.3	2.9	6.3	1.8	virginica
6.7	2.5	5.8	1.8	virginica
7.2	3.6	6.1	2.5	virginica
6.5	3.2	5.1	2	virginica
6.4	2.7	5.3	1.9	virginica
6.8	3	5.5	2.1	virginica
5.7	2.5	5	2	virginica
5.8	2.8	5.1	2.4	virginica
6.4	3.2	5.3	2.3	virginica
6.5	3	5.5	1.8	virginica
7.7	3.8	6.7	2.2	virginica
7.7	2.6	6.9	2.3	virginica
6	2.2	5	1.5	virginica
6.9	3.2	5.7	2.3	virginica
5.6	2.8	4.9	2	virginica
7.7	2.8	6.7	2	virginica
6.3	2.7	4.9	1.8	virginica
6.7	3.3	5.7	2.1	virginica
7.2	3.2	6	1.8	virginica
6.2	2.8	4.8	1.8	virginica
6.1	3	4.9	1.8	virginica

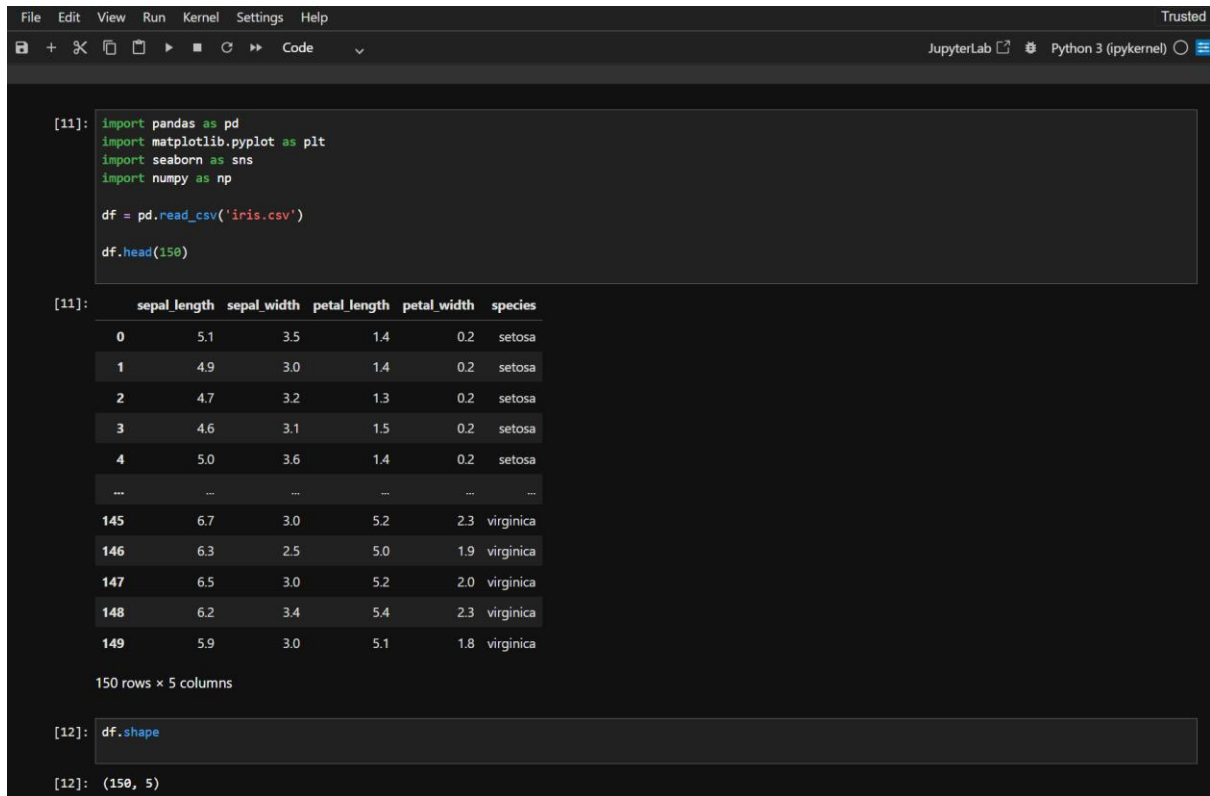
6.4	2.8	5.6	2.1	virginica
7.2	3	5.8	1.6	virginica
7.4	2.8	6.1	1.9	virginica
7.9	3.8	6.4	2	virginica
6.4	2.8	5.6	2.2	virginica
6.3	2.8	5.1	1.5	virginica
6.1	2.6	5.6	1.4	virginica
7.7	3	6.1	2.3	virginica
6.3	3.4	5.6	2.4	virginica
6.4	3.1	5.5	1.8	virginica
6	3	4.8	1.8	virginica
6.9	3.1	5.4	2.1	virginica
6.7	3.1	5.6	2.4	virginica
6.9	3.1	5.1	2.3	virginica
5.8	2.7	5.1	1.9	virginica
6.8	3.2	5.9	2.3	virginica
6.7	3.3	5.7	2.5	virginica
6.7	3	5.2	2.3	virginica
6.3	2.5	5	1.9	virginica
6.5	3	5.2	2	virginica
6.2	3.4	5.4	2.3	virginica
5.9	3	5.1	1.8	virginica

EXPERIMENT - 1

A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE, BIVARIATE AND MULTIVARIATE REGRESION

AIM

To write a Python program to implement **Univariate, Bivariate, and Multivariate Regression analysis** using the Iris dataset and to visualize the relationship between different features using graphical representations.



```
[11]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

df = pd.read_csv('iris.csv')

df.head(150)
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows x 5 columns

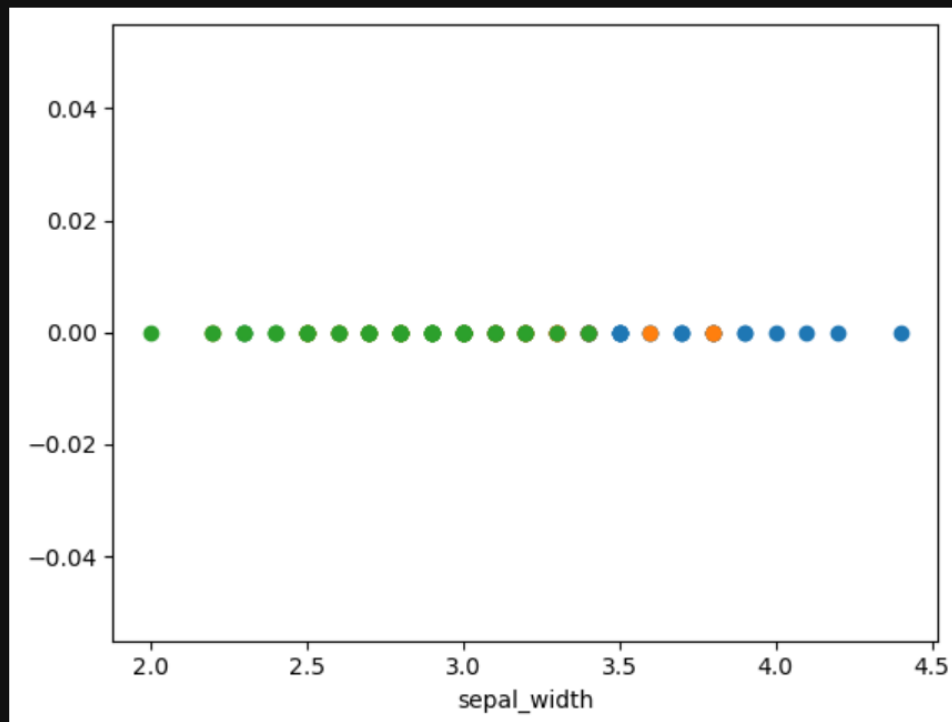
```
[12]: df.shape
```

```
[12]: (150, 5)
```

```
[13]: df.loc[df['species']=='setosa']
df_Setosa = df.loc[df['species']=='setosa']
df_Virginica = df.loc[df['species']=='virginica']
df_Versicolor = df.loc[df['species']=='versicolor']

plt.scatter(df_Setosa['sepal_width'],
            np.zeros_like(df_Setosa['sepal_width']))
plt.scatter(df_Virginica['sepal_width'],
            np.zeros_like(df_Virginica['sepal_width']))
plt.scatter(df_Versicolor['sepal_width'],
            np.zeros_like(df_Versicolor['sepal_width']))

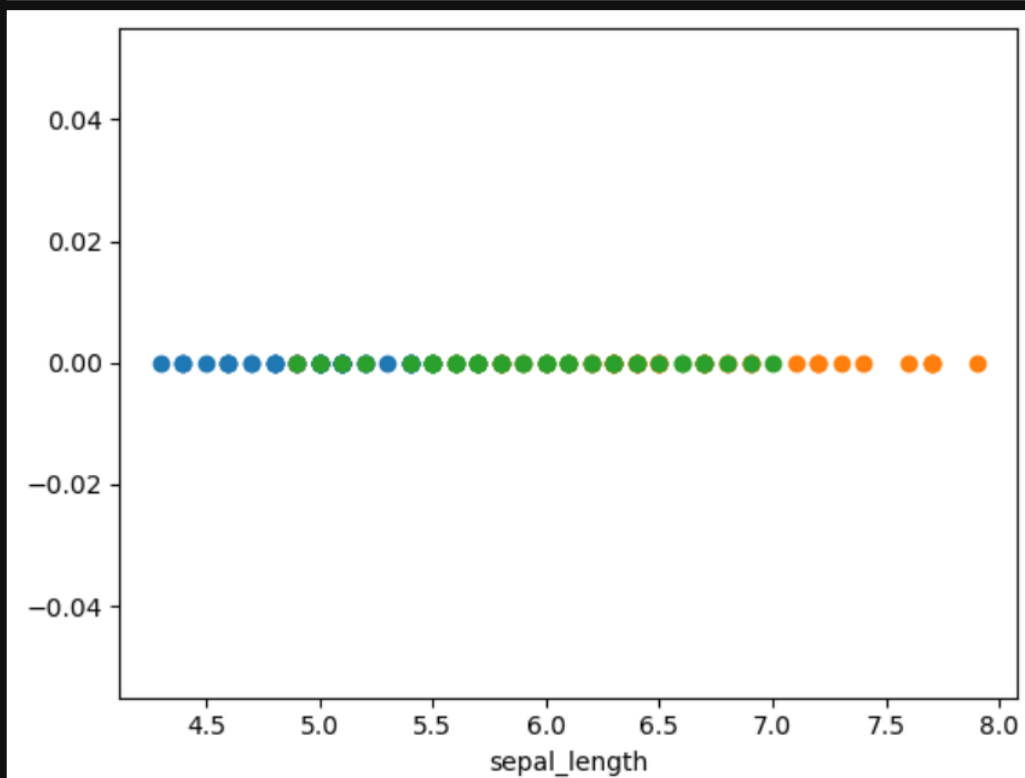
plt.xlabel('sepal_width')
plt.show()
```



```
[14]: df.loc[df['species']=='setosa']
df_Setosa = df.loc[df['species']=='setosa']
df_Virginica = df.loc[df['species']=='virginica']
df_Versicolor = df.loc[df['species']=='versicolor']

plt.scatter(df_Setosa['sepal_length'],
            np.zeros_like(df_Setosa['sepal_length']))
plt.scatter(df_Virginica['sepal_length'],
            np.zeros_like(df_Virginica['sepal_length']))
plt.scatter(df_Versicolor['sepal_length'],
            np.zeros_like(df_Versicolor['sepal_length']))

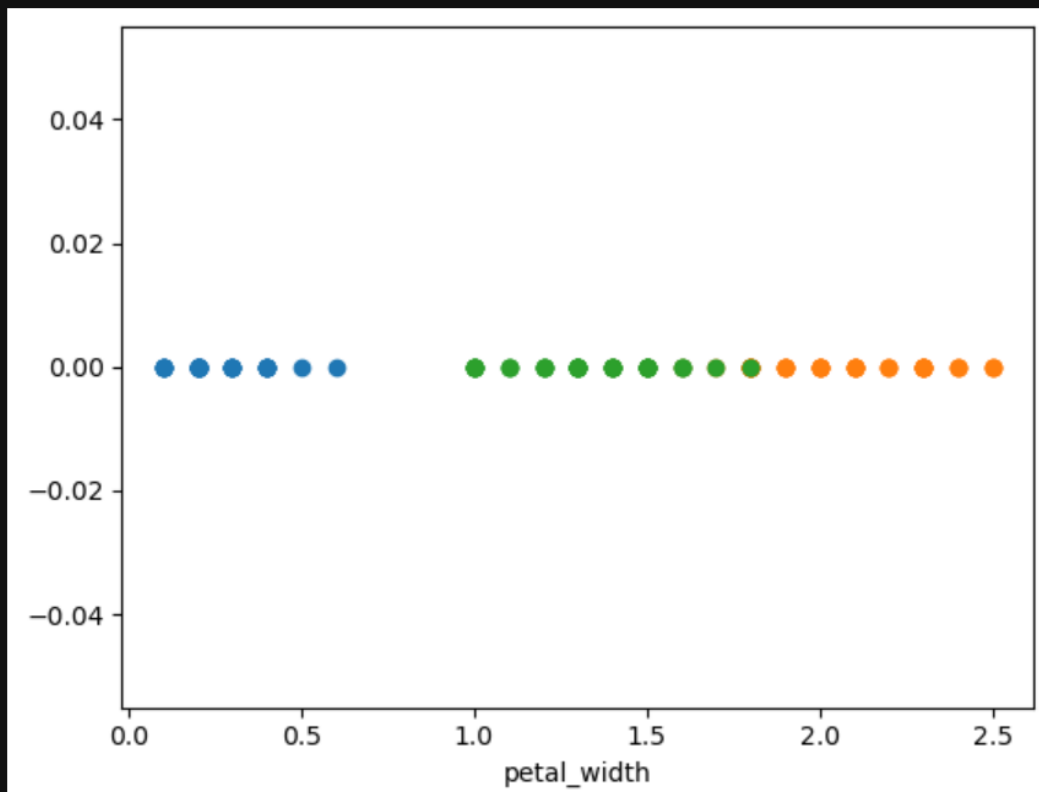
plt.xlabel('sepal_length')
plt.show()
```



```
[15]: df.loc[df['species']=='setosa']
df_Setosa = df.loc[df['species']=='setosa']
df_Virginica = df.loc[df['species']=='virginica']
df_Versicolor = df.loc[df['species']=='versicolor']

plt.scatter(df_Setosa['petal_width'],
            np.zeros_like(df_Setosa['petal_width']))
plt.scatter(df_Virginica['petal_width'],
            np.zeros_like(df_Virginica['petal_width']))
plt.scatter(df_Versicolor['petal_width'],
            np.zeros_like(df_Versicolor['petal_width']))

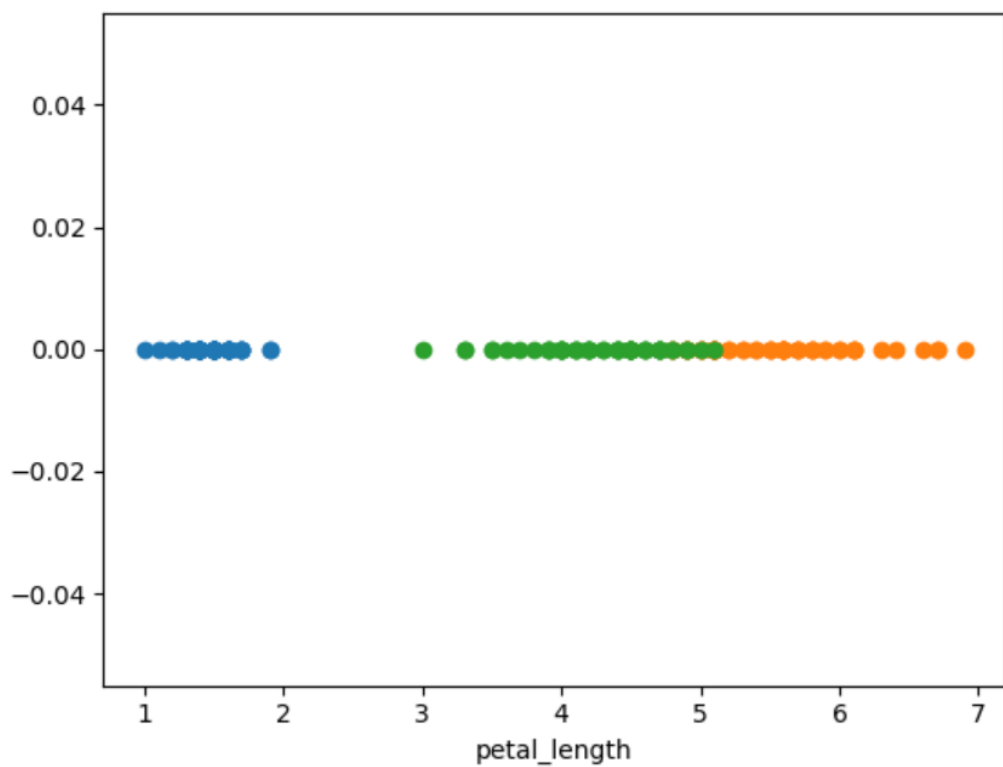
plt.xlabel('petal_width')
plt.show()
```



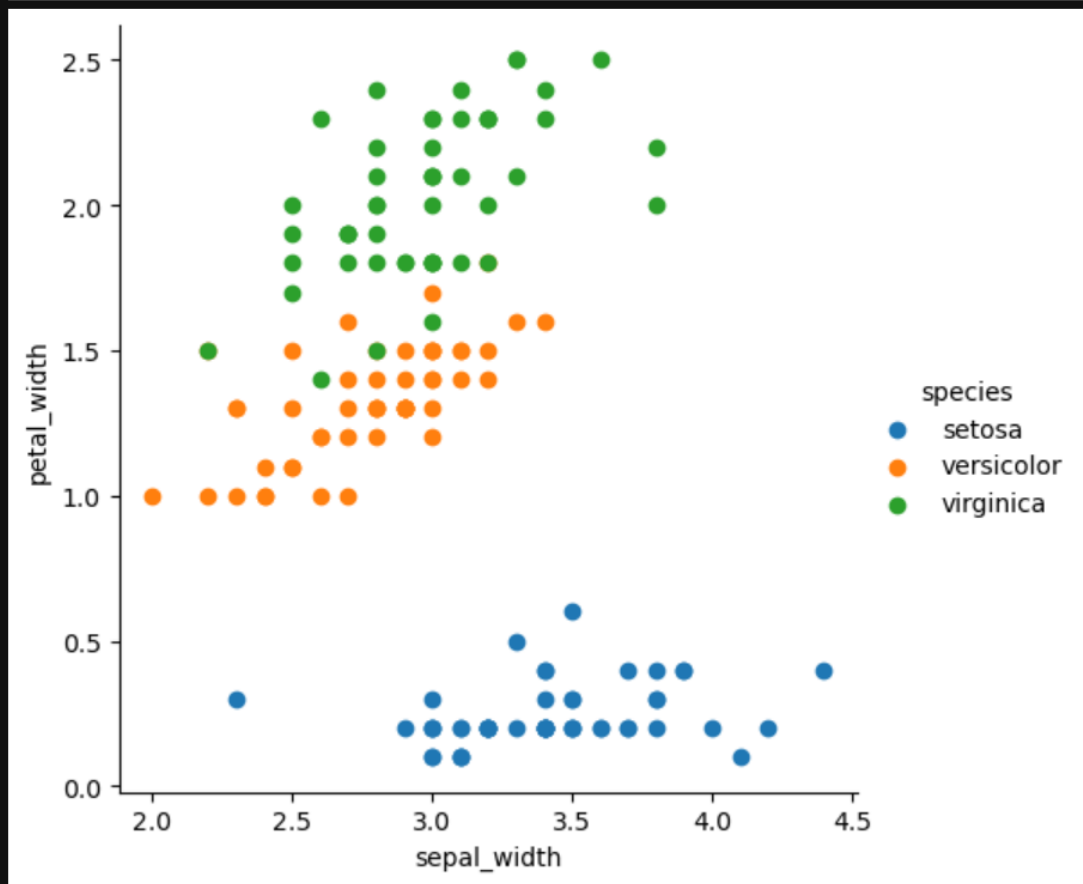

```
[16]: df.loc[df['species']=='setosa']
df_Setosa = df.loc[df['species']=='setosa']
df_Virginica = df.loc[df['species']=='virginica']
df_Versicolor = df.loc[df['species']=='versicolor']

plt.scatter(df_Setosa['petal_length'],
            np.zeros_like(df_Setosa['petal_length']))
plt.scatter(df_Virginica['petal_length'],
            np.zeros_like(df_Virginica['petal_length']))
plt.scatter(df_Versicolor['petal_length'],
            np.zeros_like(df_Versicolor['petal_length']))

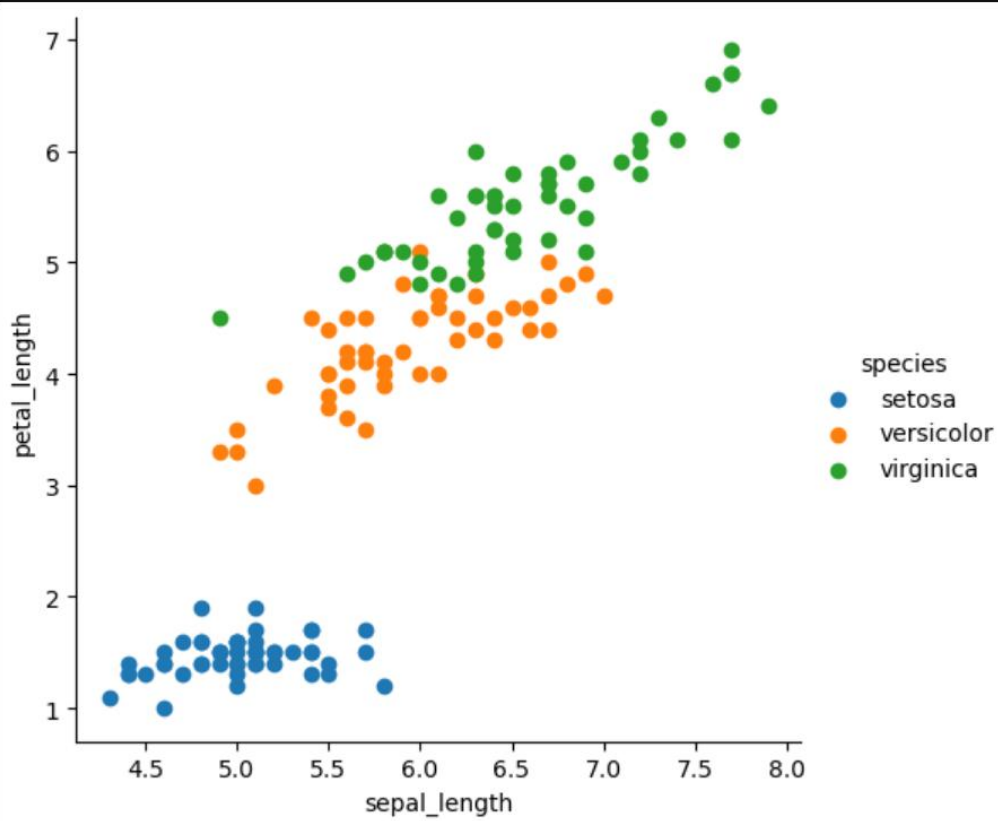
plt.xlabel('petal_length')
plt.show()
```



```
[17]: sns.FacetGrid(df, hue='species', height=5)\
      .map(plt.scatter, "sepal_width", "petal_width")\
      .add_legend()\
      plt.show()
```

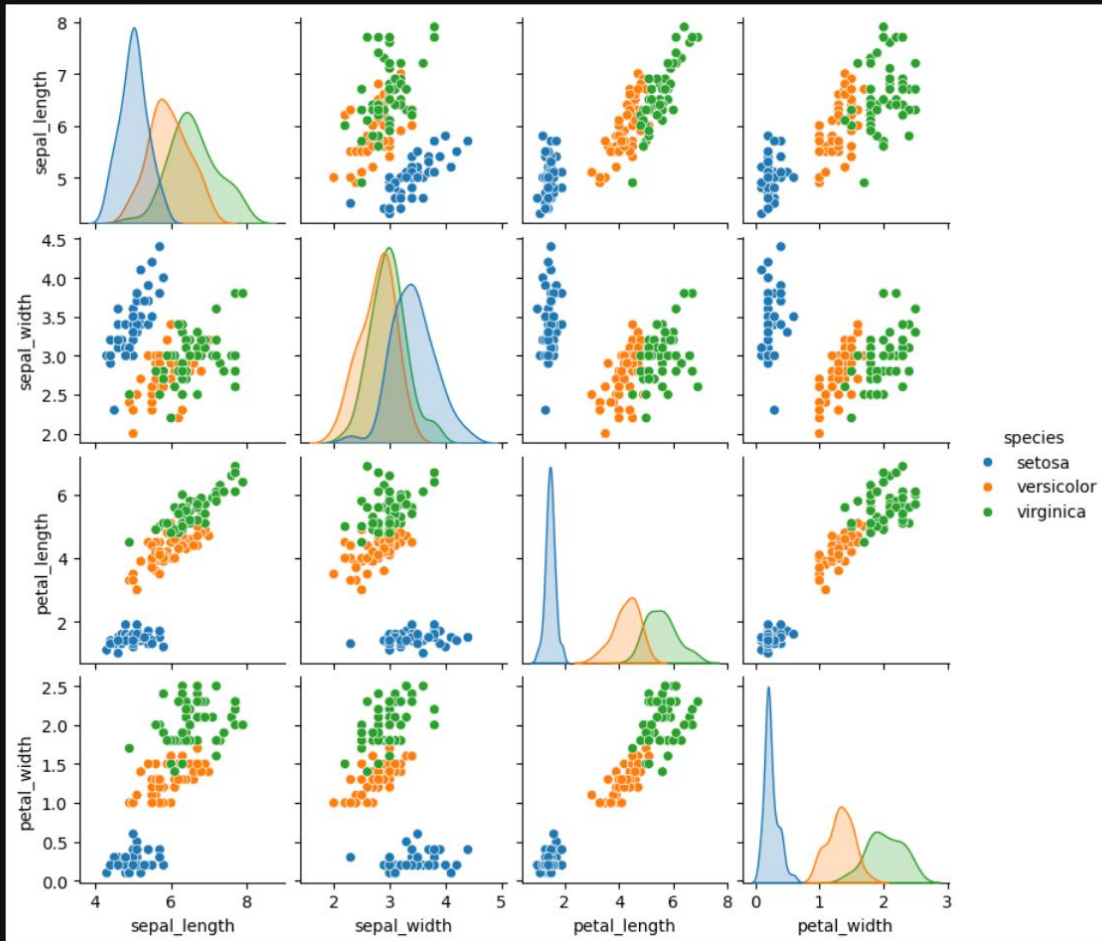


```
[18]: sns.FacetGrid(df, hue='species', height=5)\n      .map(plt.scatter, "sepal_length", "petal_length")\n      .add_legend()\n\nplt.show()
```



```
[19]: sns.pairplot(df, hue="species", height=2)
```

```
[19]: <seaborn.axisgrid.PairGrid at 0x2582f70f770>
```



RESULT

Thus, the Python program for implementing **Univariate, Bivariate, and Multivariate regression analysis** was successfully executed using the Iris dataset. The relationships between different features were visualized using scatter plots and pair plots, and the distribution and correlation among the variables were observed successfully.